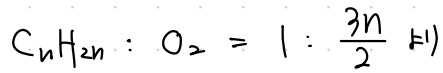
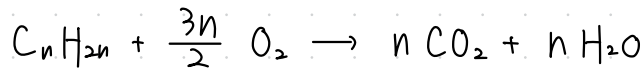


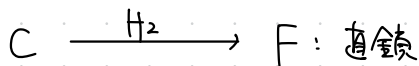
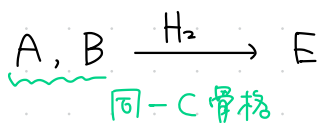
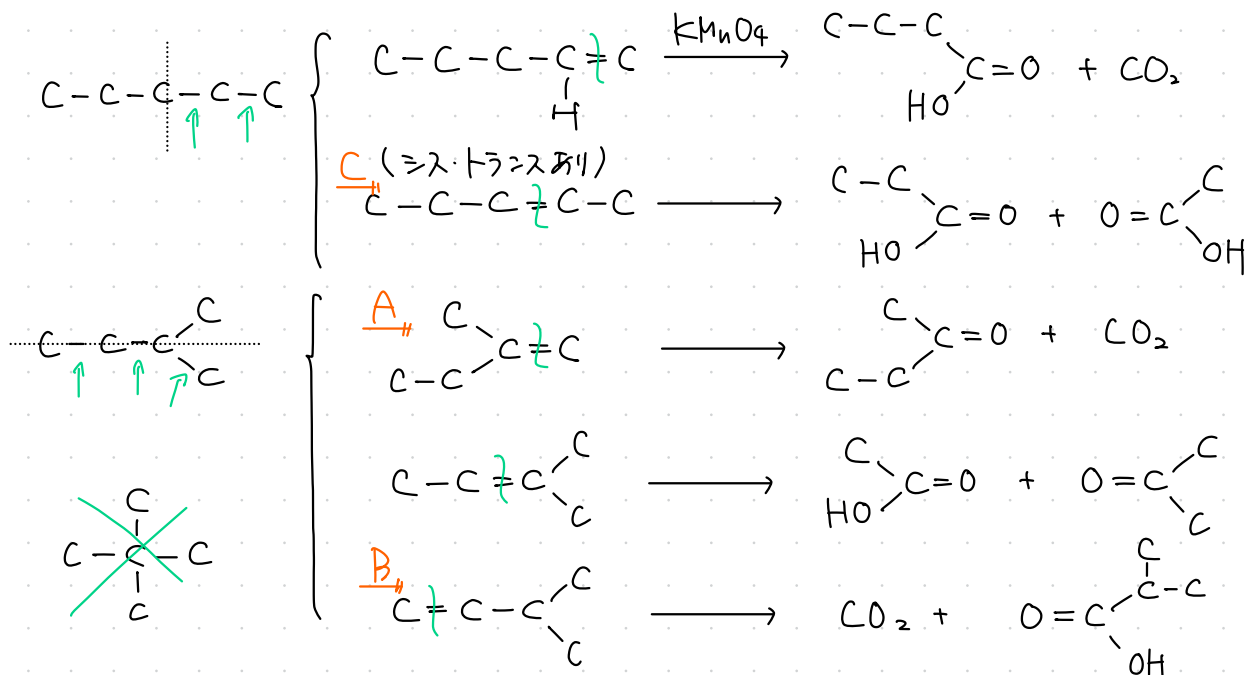
2-1

問1 (a), (b) は) A, B, C はアルケン C_nH_{2n}

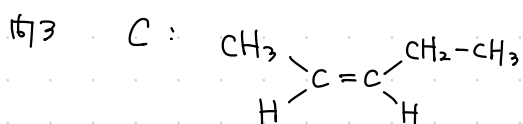
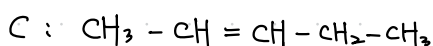
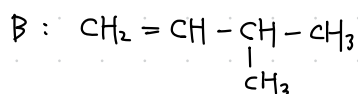
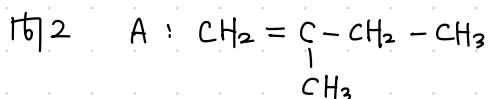


$$1 \text{ mol} \times \frac{3n}{2} = \frac{168 \text{ L}}{22.4 \text{ L/mol}} \therefore n = 5 \quad \underline{C_5H_{10}}$$

問5 C_5H_{10} $U = 1$ $\therefore C=C \times 1$



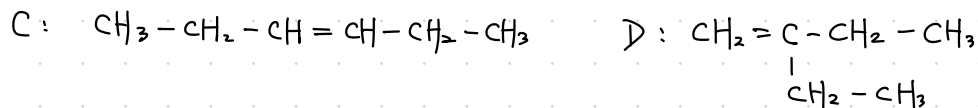
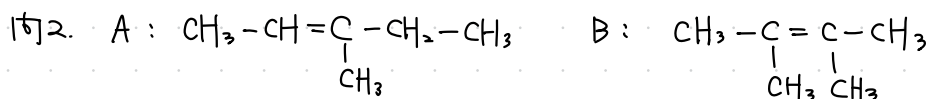
問1 C_5H_{10}



沸点 $\rightarrow$ 分子間力 $\left\{ \begin{array}{l} \text{水素結合} \\ \text{ファンデルワールス力} \end{array} \right.$
 $\Rightarrow$ ス型の方が極性 $\odot \rightarrow$ 分子間力 $\odot$

2-2

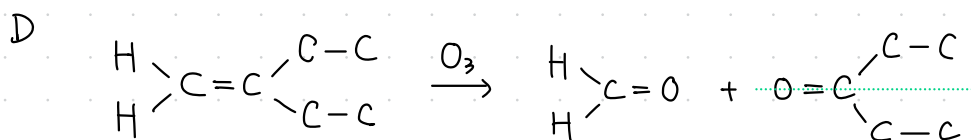
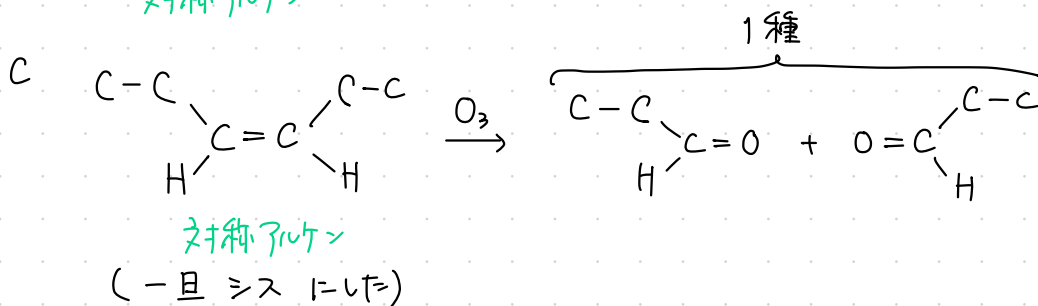
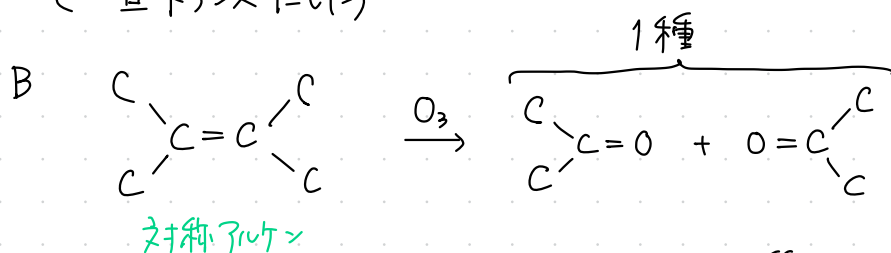
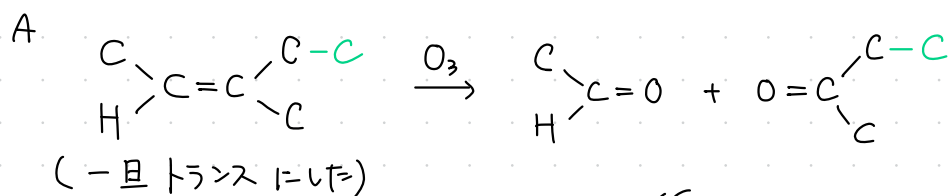
問1 C_6H_{12}



問3 $A \geq C$



$$\frac{2.1 \text{ g}}{14n \text{ g/mol}} = \frac{4.0 \text{ g}}{160 \text{ g/mol}} \quad \therefore n=6 \quad \underline{C_6H_{12}}$$



2-3

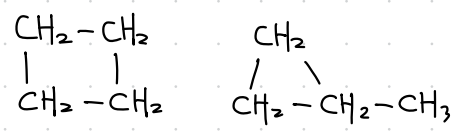
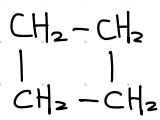
問1 A $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}_3$ 1-ブテン

B $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$ 2-ブテン

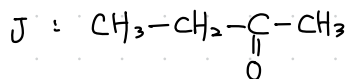
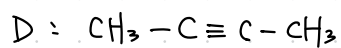
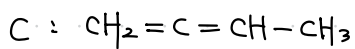
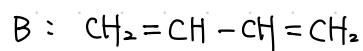
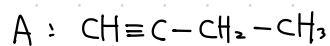
C $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_3$ ブタン

問2 D: 2 E: 3 (メソ体あり)

問3

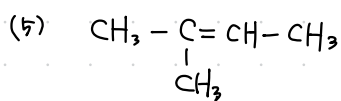
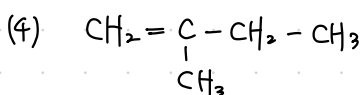
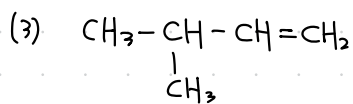


2 - 4



2 - 5

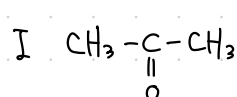
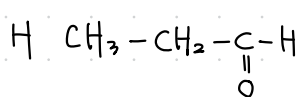
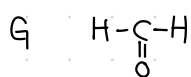
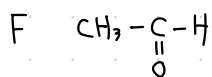
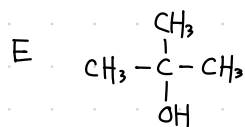
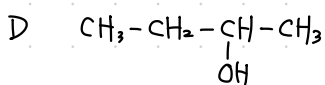
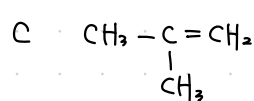
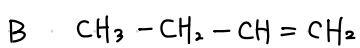
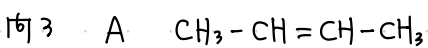
(1) 5 (2) 1



2 - 6

161 102

162 C_4H_8



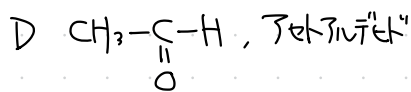
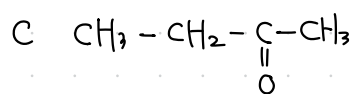
164 F

2 - [7]

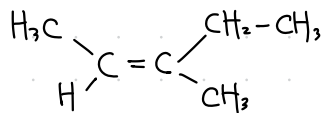
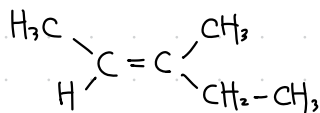
問1 略

問2 C C_4H_8O D C_2H_4O

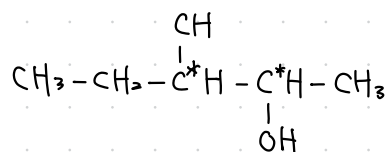
問3



問4.



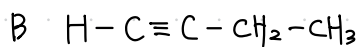
問5



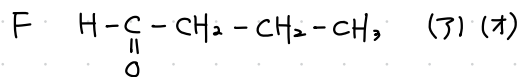
2 - [8]

問1 アセチレン, γ -ブチレン

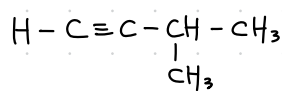
問2 (1) A $CH_3-C \equiv C-CH_3$



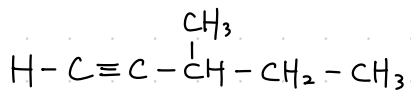
(2) E $CH_3-\overset{\overset{O}{\parallel}}{C}-CH_2-CH_3$ (イ)



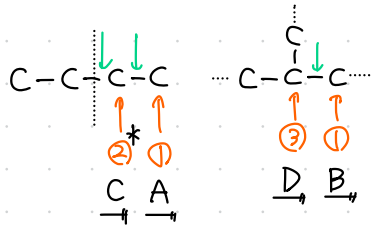
問3



問4.

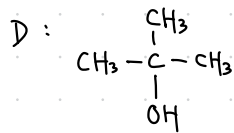
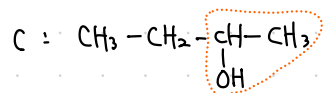
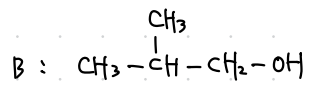
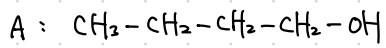


3-11 $C_4H_{10}O$ $U=1$ $\left\{ \frac{\text{アルコール}}{\text{エーテル}} \right\}$



アルコール $A^{\textcircled{1}}$ $B^{\textcircled{1}}$ C^* $D^{\textcircled{2}}$
 O_x E F G X
 アルコール アルコール
 b.p. $A > B$

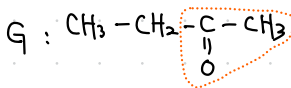
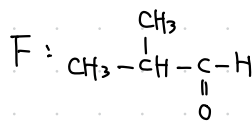
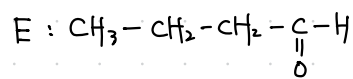
問1



問2

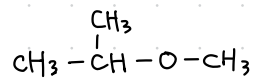
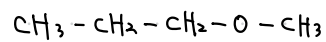
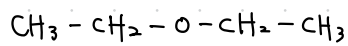
ヒドロキシ基を持ち、
分子間での水素結合を形成するため。

問3

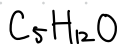


C, G

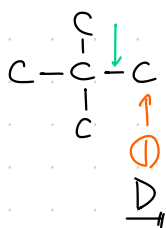
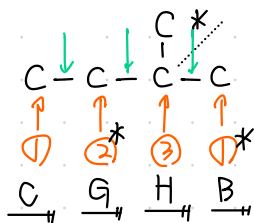
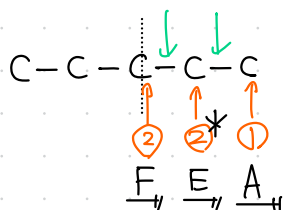
問4



3 - 2

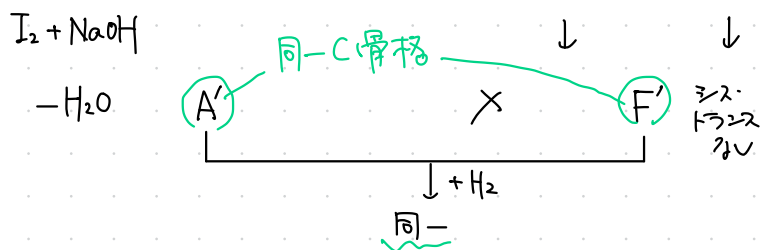


$$U=0 \quad \left\{ \begin{array}{l} \gamma_{\mu\nu} = -\gamma_{\nu\mu} \\ I = -\bar{I} \end{array} \right.$$



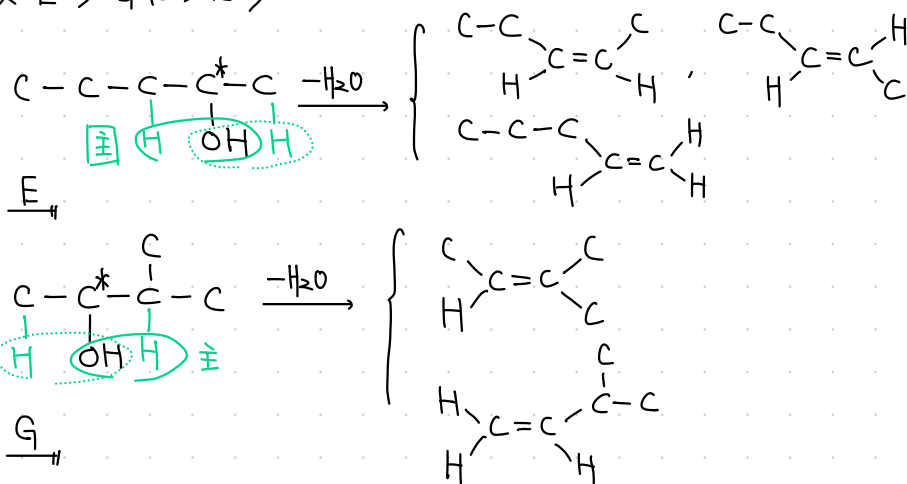
アルコール ① A ① B^{*} ① C ① D ② E^{*} ② F ③ G^{*} ③ H

Ox { アルデヒド } { ケトン } X



$\left. \begin{array}{l} \text{「脱水した} \sim \text{」} \\ \text{「H}_2\text{付加した} \sim \text{」} \end{array} \right\} \text{同一化合物}$
 $\longrightarrow \text{同一C骨格}$

$\langle 2 \text{級 } E^*, G^* | \Rightarrow WZ \rangle$

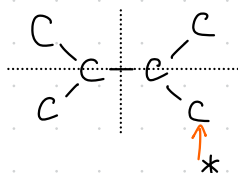
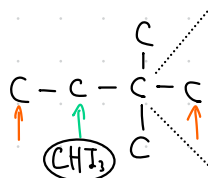
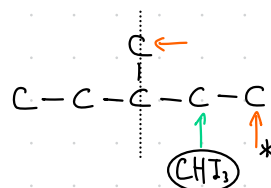
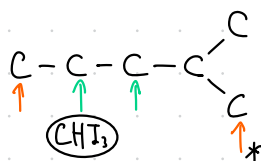
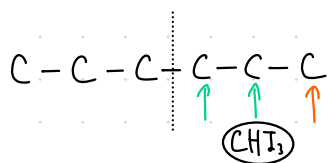


(2) $\xrightarrow{6}$ (立体异构: 7)

3 - ③

← : =O (アルデヒド)
← : =O (ケト>)

問1

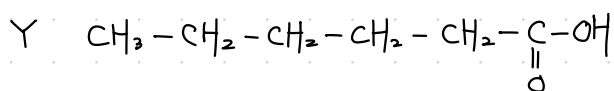
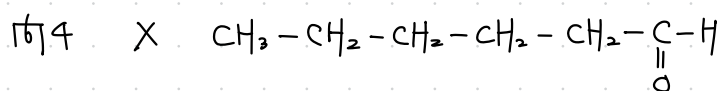
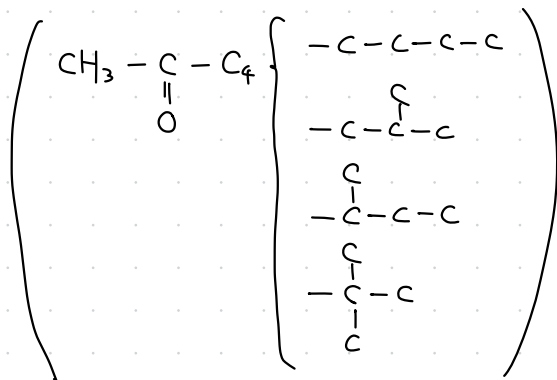


(a) 8

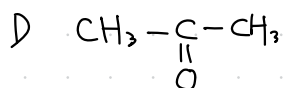
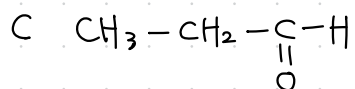
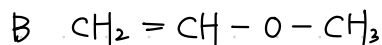
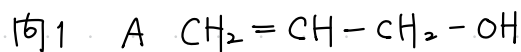
(b) 6

問2. 3

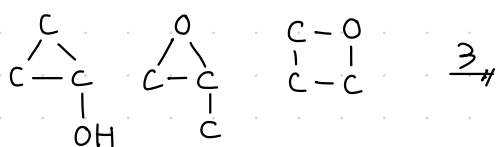
問3. 4

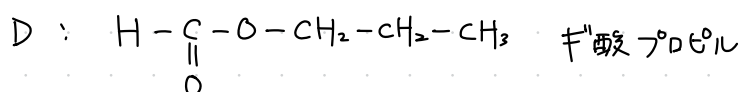
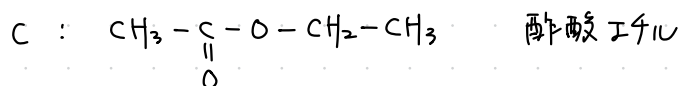
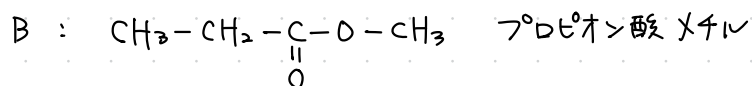
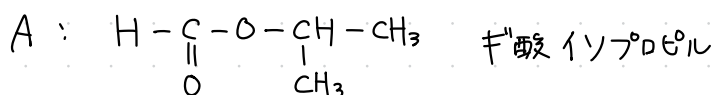
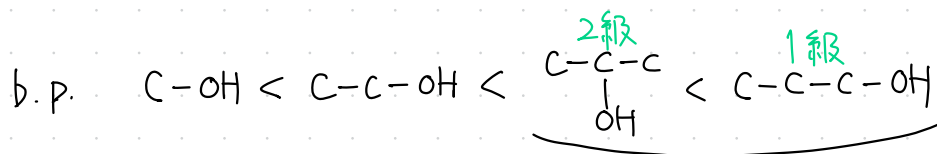
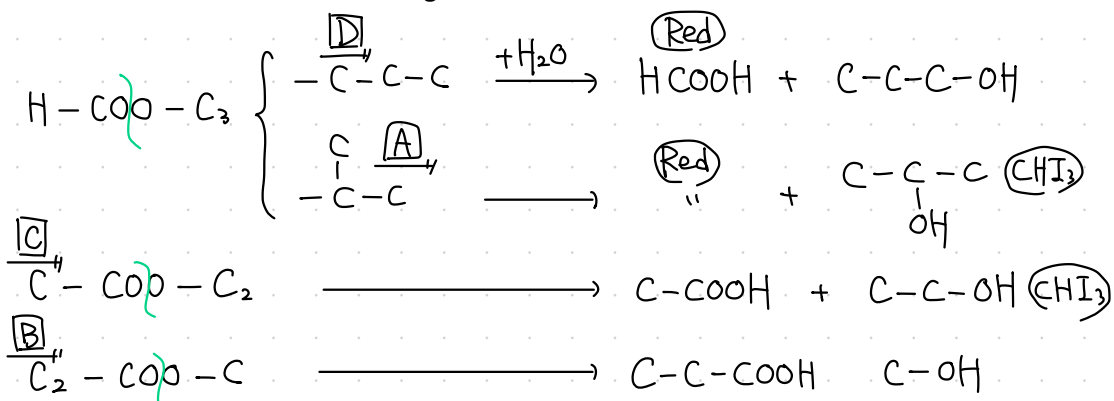
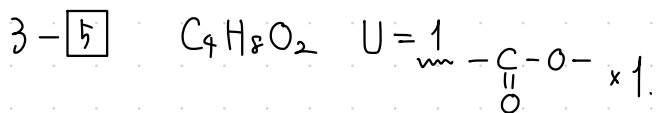


3 - ④



問2





3-6

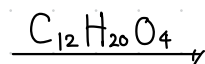
$$\text{例1} \quad w_C = 264 \text{ g} \times \frac{C(12)}{CO_2(44)} = 72 \text{ g}$$

$$w_H = 90 \text{ g} \times \frac{2H(2)}{H_2O(18)} = 10 \text{ g}$$

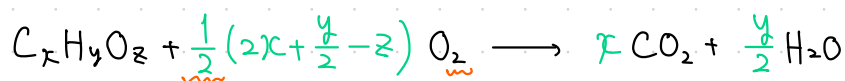
$$w_O = 114 - (72 + 10) = 32 \text{ g}$$

$$C : H : O = \frac{72 \text{ g}}{12 \text{ g/mol}} : \frac{10 \text{ g}}{1.0 \text{ g/mol}} : \frac{32 \text{ g}}{16 \text{ g/mol}} = 3 : 5 : 1$$

$$(C_3H_5O)_n = 57n = 228 \quad n = 4$$



<別解> 燃烧例, 分子量既知の時



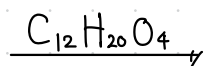
$$\begin{matrix} C_xH_yO_z : CO_2 = 1 : x \text{ mol} \\ (288) \quad \quad (44) \end{matrix}$$

$$\frac{114 \text{ g}}{288 \text{ g/mol}} \times x = \frac{264 \text{ g}}{44 \text{ g/mol}} \times 1 \quad \therefore x = 12$$

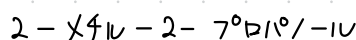
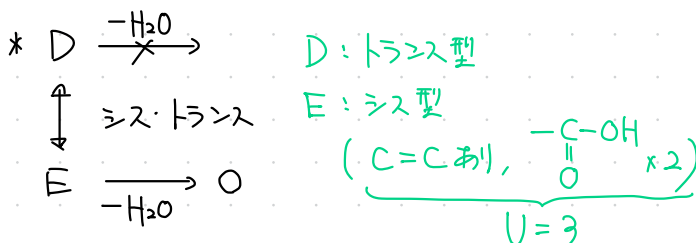
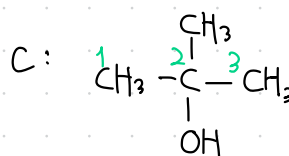
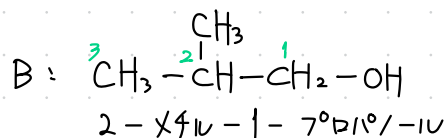
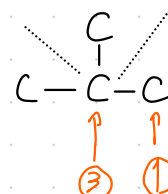
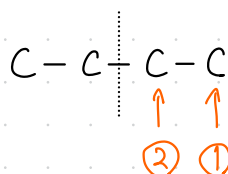
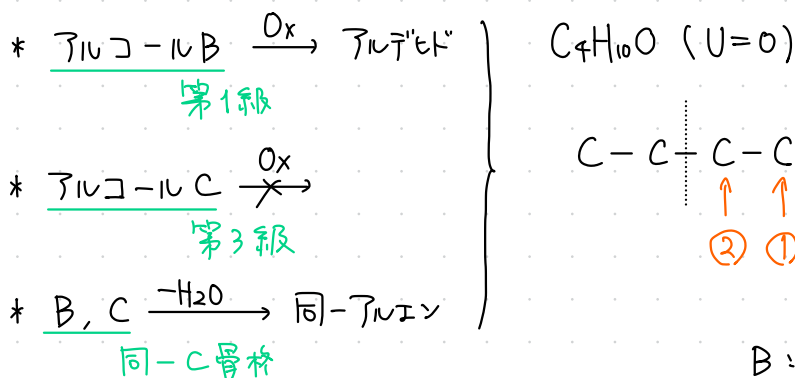
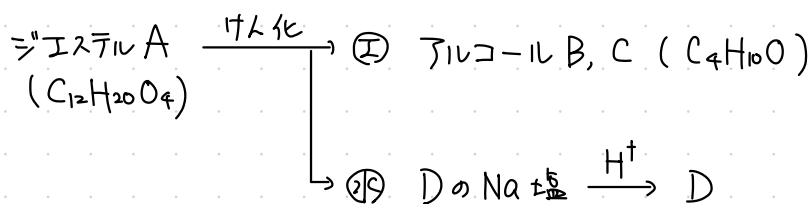
$$\begin{matrix} C_xH_yO_z : H_2O = 1 : \frac{y}{2} \text{ mol} \\ (288) \end{matrix}$$

$$\frac{114 \text{ g}}{288 \text{ g/mol}} \times \frac{y}{2} = \frac{90 \text{ g}}{18 \text{ g/mol}} \times 1 \quad \therefore y = 20$$

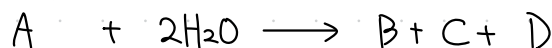
$$C_{12}H_{20}O_z = 164 + 16z = 228 \quad \therefore z = 4$$



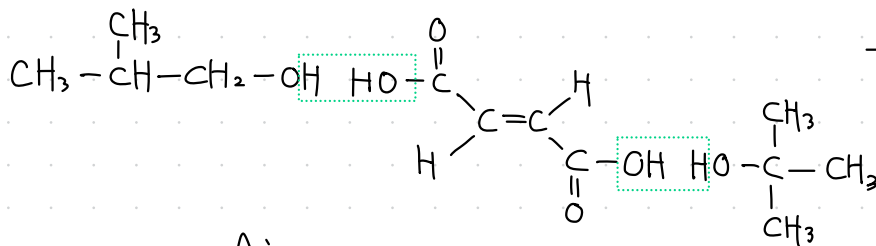
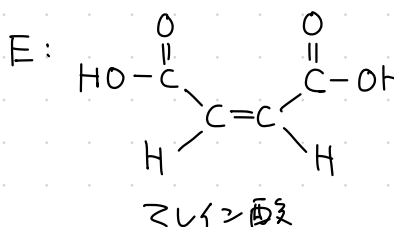
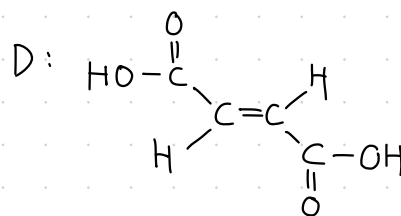
問2. A: $C_{12}H_{20}O_4$ $U = \frac{2 \times 12 + 2 - 20}{2} = 3$ C=C $\times 1$ + $-\overset{\overset{O}{\parallel}}{C}-O- \times 2$



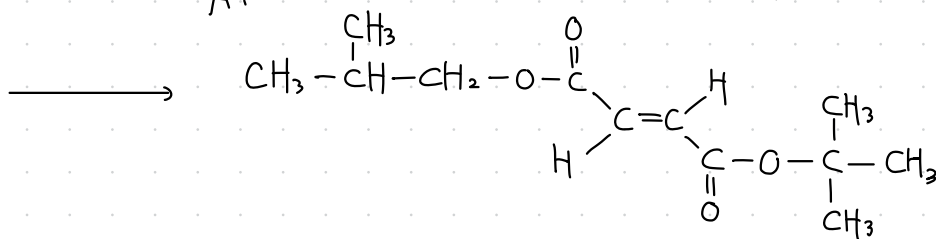
ジエステル



$\therefore D = A + 2H_2O - (B + C)$
 $= C_{12}H_{20}O_4 + 2H_2O - 2 \times C_4H_{10}O$
 $= C_4H_4O_4$ (114)
 ($U=3$)



A:



3 - [7]

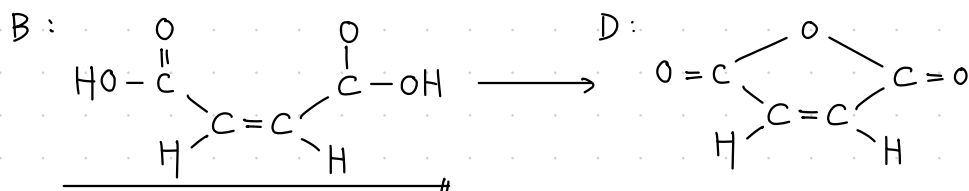
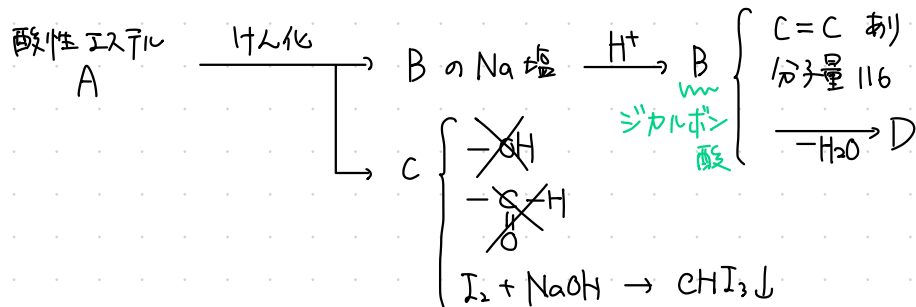
問1 $C_7H_8O_4$

$$C:H:O = \frac{53.8}{12} : \frac{5.1}{1.0} : \frac{41.1}{16} = 4.48 : 5.1 : \frac{2.56}{\text{①}}$$

$$\div 1.75 : 1.99 : 1 = 7:8:4$$

$$130 \leq (C_7H_8O_4)_n \leq 170 \quad \therefore n=1 \quad \underline{C_7H_8O_4} \quad (U=4)$$

問2

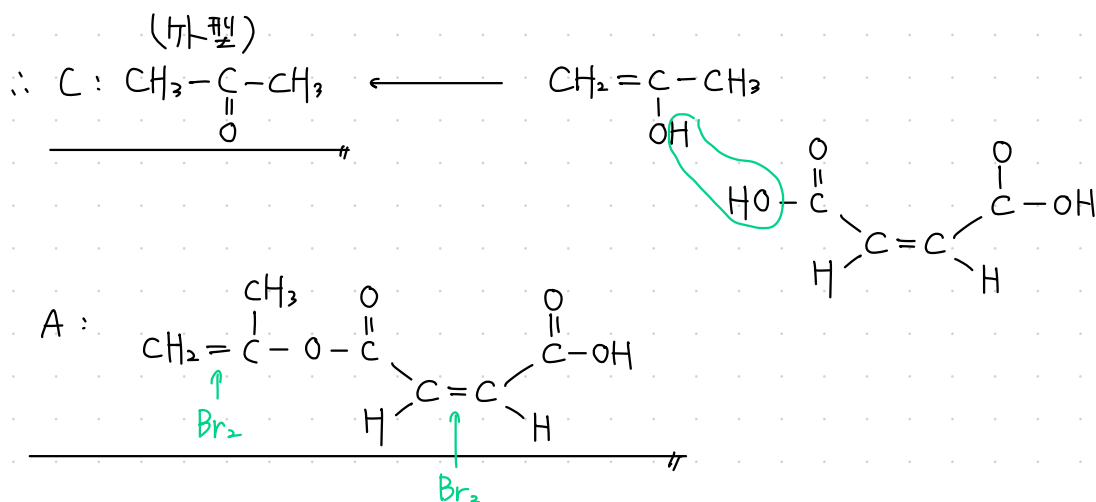


$$A + H_2O \longrightarrow B + C \quad \therefore C = A + H_2O - B$$

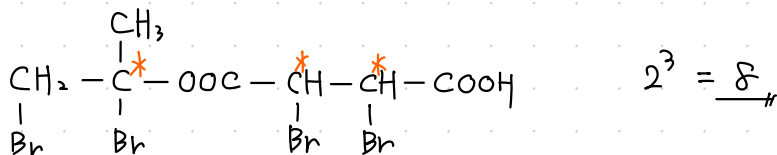
$$= C_7H_8O_4 + H_2O - C_4H_4O_4$$

$$= \underline{C_3H_6O} \quad (U=1)$$

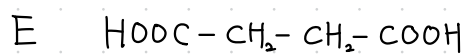
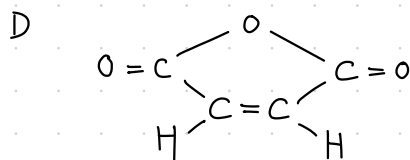
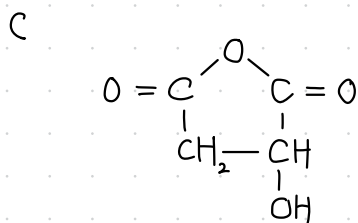
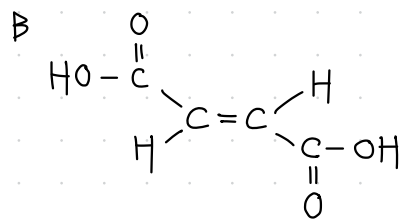
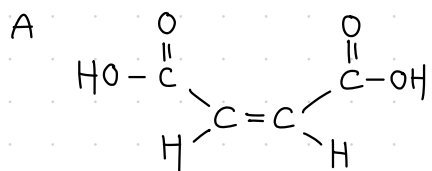
エステルの加水分解生酸性物として $-OH$ 基を持つものがない
 $\rightarrow$ エノール型 か ケト型 に変わった。



問3



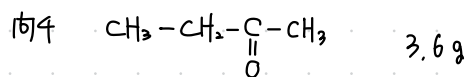
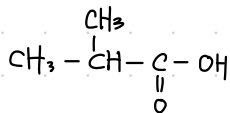
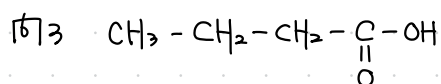
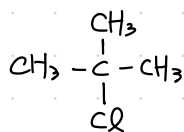
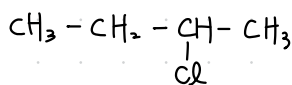
3-8



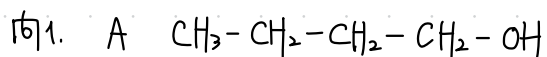
3-9

1 4

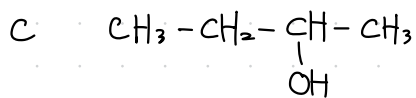
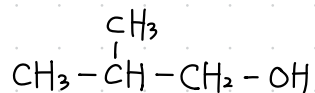
2.



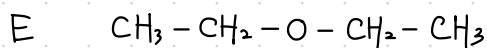
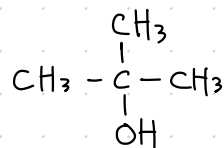
3-10



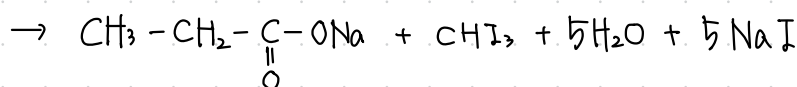
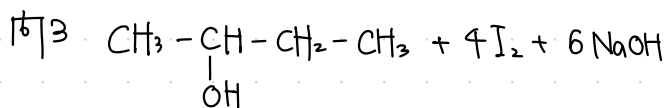
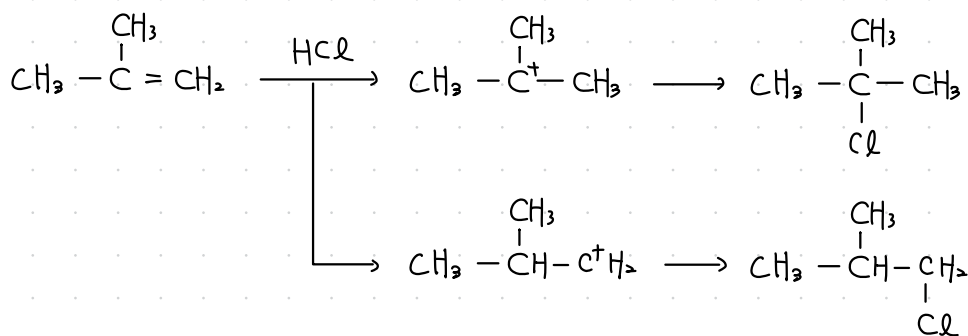
B



D



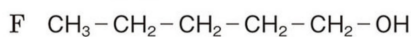
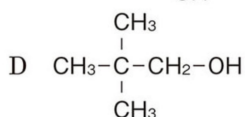
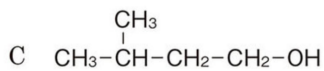
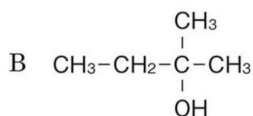
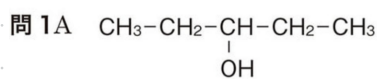
問2



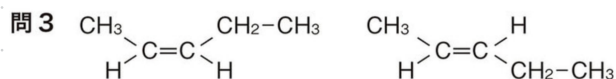
問4 C

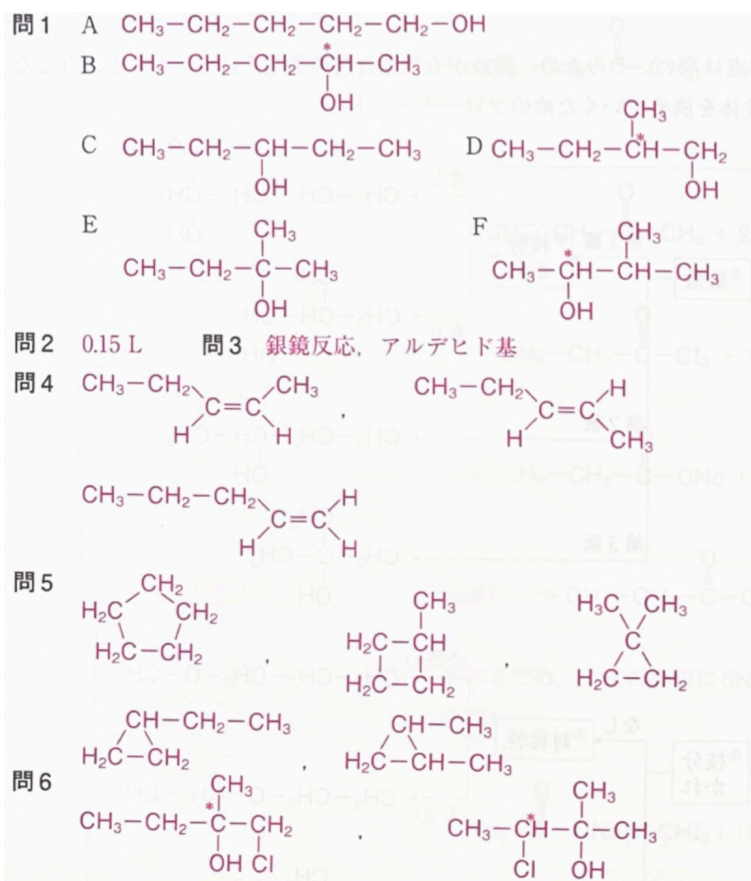
問5 エーテル系結合

3-11



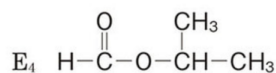
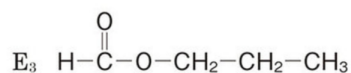
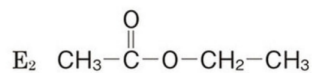
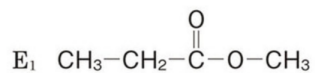
問2 K



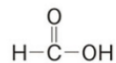


3-13

問1



問2 ギ酸は分子内にホルミル基を持つため。(19字)

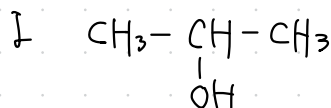
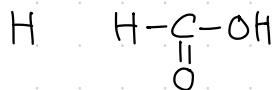
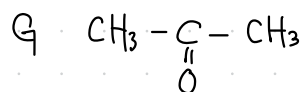
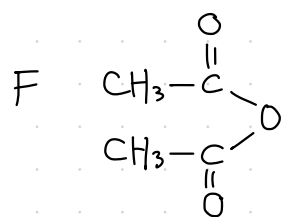
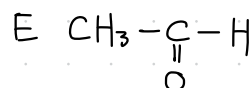
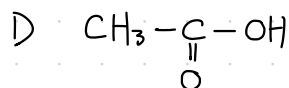
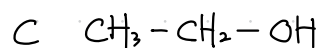
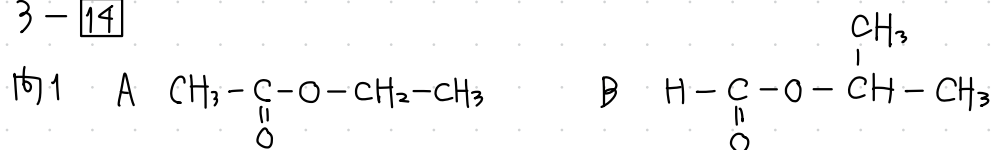


問3 酢酸カルシウム



問4 4.4 g

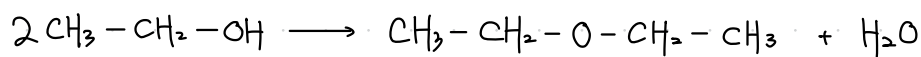
3-14



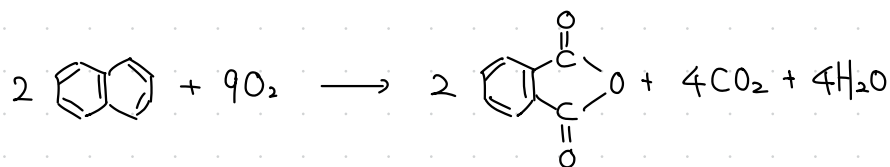
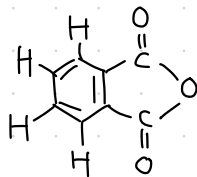
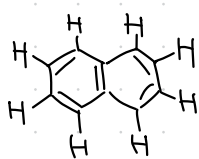
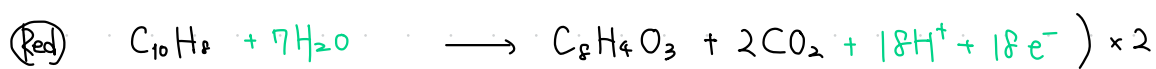
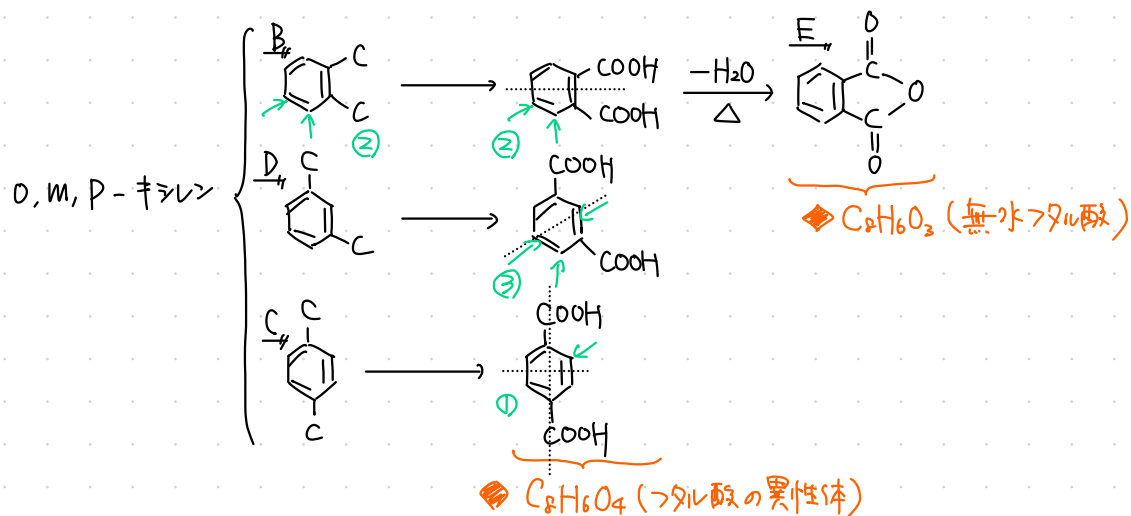
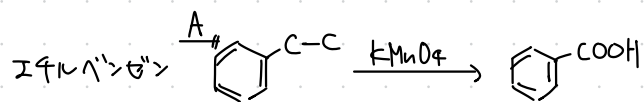
問2,3 略

問4. テトラアルデヒド

問5 シエチルエーテル



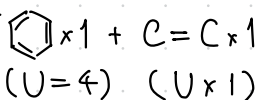
4-① C_8H_{10} $U = 4$  $\times 1$



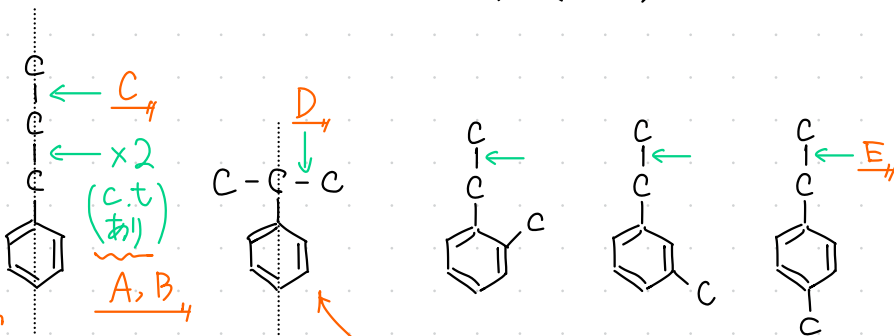
4-2

C_9H_{10}

$U = 5$

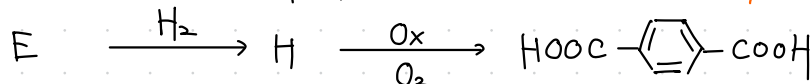
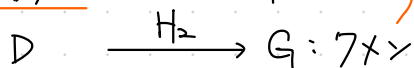


内2.



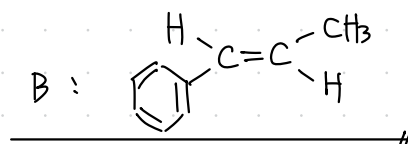
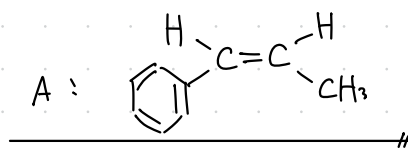
< 1置換体 >

< 2置換体 >



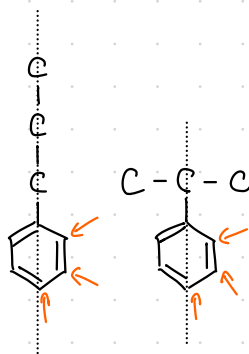
A, B: シス・トランス異性体

融点 $A < B$
シス トランス



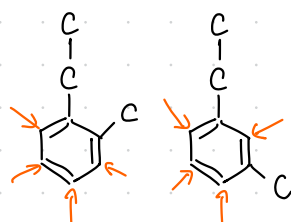
内1. 8

内3. 22



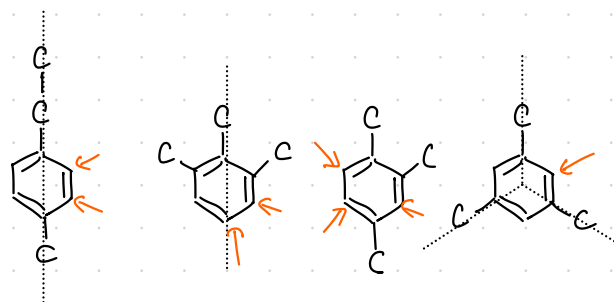
< 1置換体 >

Cl置換: 6



< 2置換体 >

Cl置換: 10



< 3置換体 >

Cl置換: 6

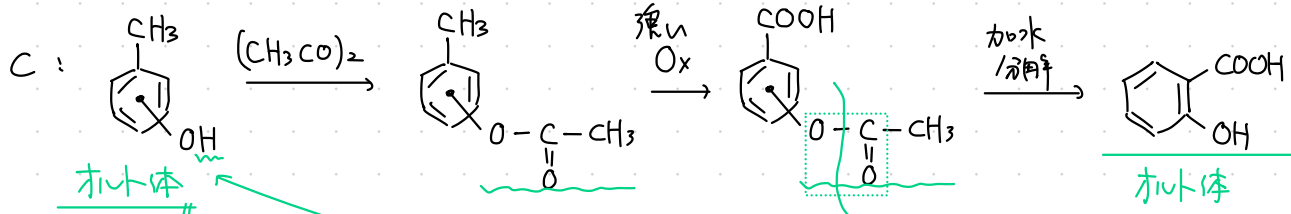
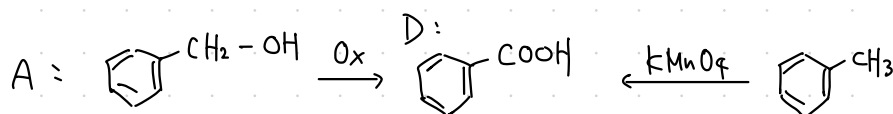
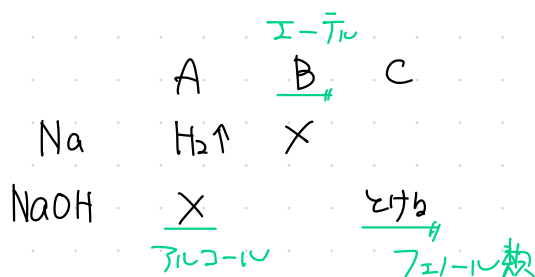
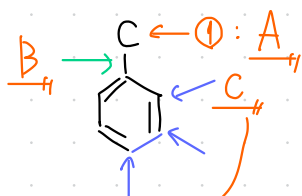
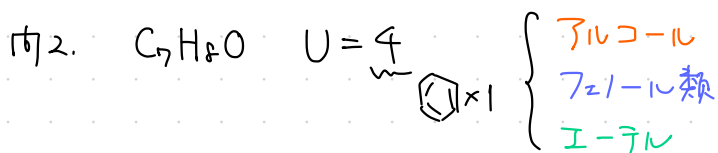
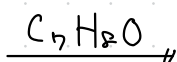
4-③

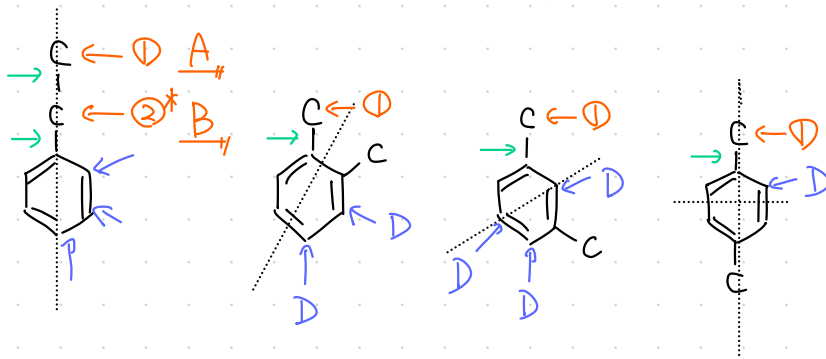
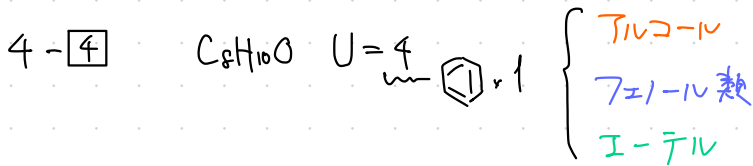


条件より $l : \frac{m}{2} = 7 : 4 \quad \therefore (l, m) = (7, 8)$

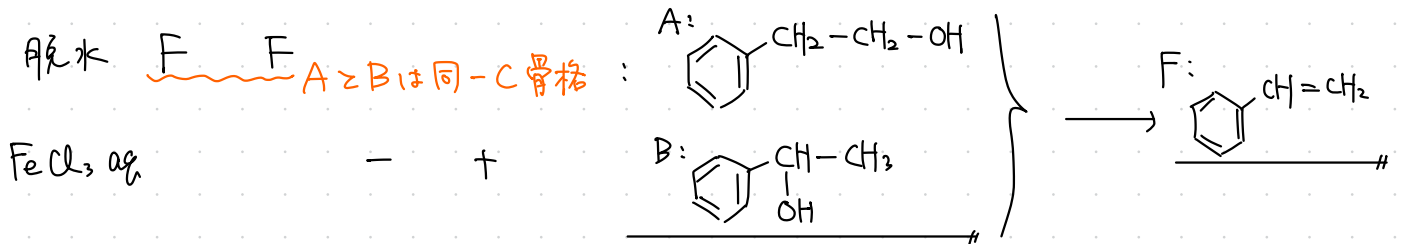
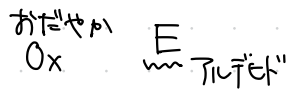
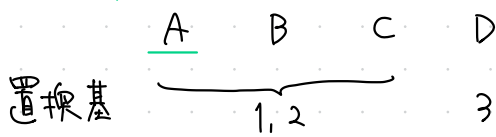
$\therefore (C_7H_8O_n)_k = (92 + 16n)k = 108$

$\therefore (n, k) = (1, 1)$

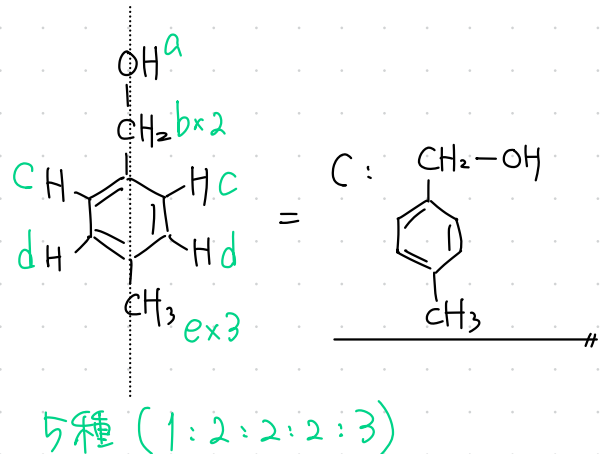
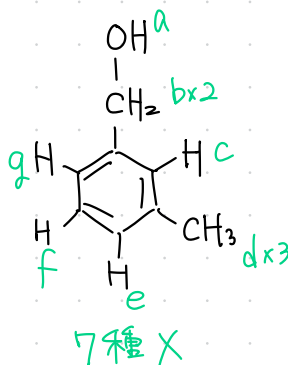
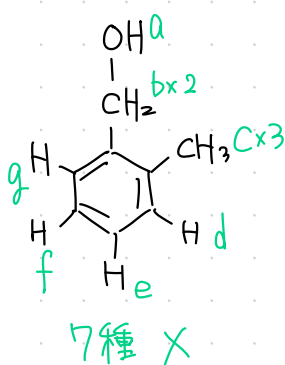




第1級
アルコール



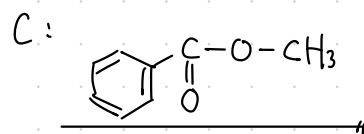
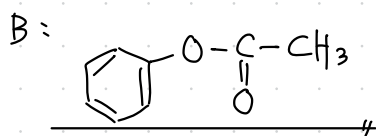
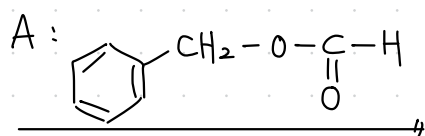
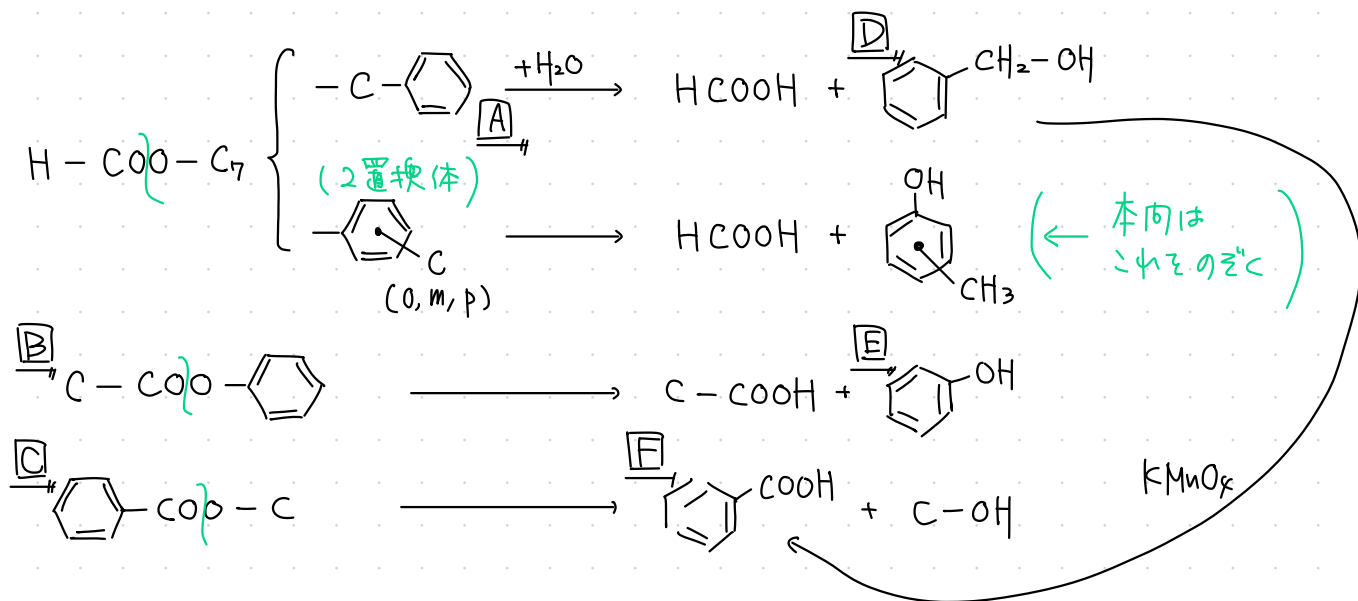
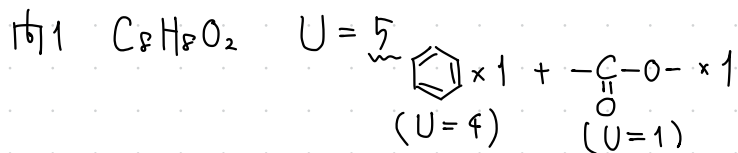
C : 1 or 2 置換体のアルコール



内2

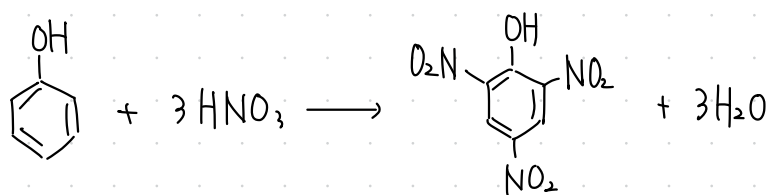
D : フェニル類 (3 置換体), $\underline{6}$

4-7



162. E

163



ヒョウリ=酸

2, 4, 6 - トリトロフエノール